GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

# Relationships in Triangles

*18 points* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Part 1. Triangle Inequality and the Converse

*3 points MA.8.GR.1.3*

**1.** Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ● **C. 6 cm, 8 cm, 13 cm**
- ○ D. 3 cm, 10 cm, 15 cm

***C***

*The two shorter sides against the longest: 6 + 8 = 14, more than 13. A lies flat; B and D pass only when the wrong pair is added.*

*Scoring: 1 point.*

**2.** Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☒ **A. 17, 8, 15**
- ☐ B. 5, 6, 7
- ☒ **C. 20, 21, 29**
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

***A and C***

*Square all three and test the longest as c: 64 + 225 = 289 = $17^2$ (the longest listed first) and 400 + 441 = 841 = $29^2$. Both and nothing else for the point.*

*Scoring: 1 point.*

**3.** Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ● **B.** $9 < x < 21$
- ○ C. $9 \le x \le 21$
- ○ D. $0 < x < 21$

***B***

*15 − 6 = 9 and 15 + 6 = 21, never equal to either.*

*Scoring: 1 point.*

## Part 2. The Pythagorean Theorem

*6 points MA.8.GR.1.1*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**4.** Find the missing side of each right triangle.

**a.** Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

***15 in***

*$c^2 = 81 + 144 = 225$*

**b.** Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

***11 m***

*a leg: subtract — 3,721 − 3,600 = 121. Adding the squares gives a 'leg' longer than the hypotenuse.*

*Scoring: a 1 · b 1 = 2 points.*

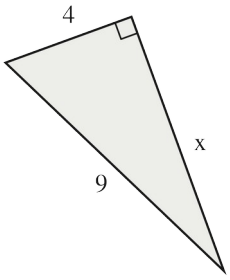
**5.** A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

***14.8 cm (exactly √218 cm)***

*$c^2 = 49 + 169 = 218$; √218 = 14.76…*

*Scoring: 1 point.*

**6.** Which equation can be used to find x? *1 pt*




- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 2 \cdot 9$
- ● **D.** $4^2 + x^2 = 9^2$

***D***

*The 9 is across from the right angle, so it is c and stands alone: $x^2 = 81 − 16 = 65$.*

*Scoring: 1 point.*

**7.** A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

**a.** How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

***13.5 m  (exactly √181 m)***

*the path is the hypotenuse: 81 + 100 = 181*

**b.** How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

***5.5 m***

*9 + 10 = 19 m along two sides; 19 − 13.5. Follow-through on part (a).*

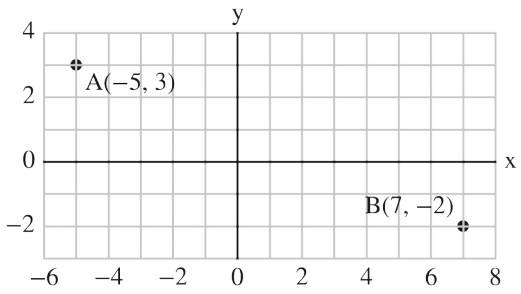
*Scoring: a 1 · b 1 = 2 points.*

## Part 3. Distance on the Coordinate Plane

*6 points  MA.8.GR.1.2*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**8.** Find the distance between points A and B. *1 pt*




***13 units***

*Across 7 − (−5) = 12, down 3 − (−2) = 5: $c^2$ = 144 + 25 = 169.*

*Scoring: 1 point.*

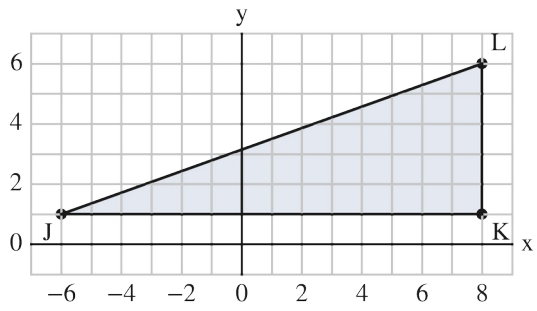
**9.** Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

***5.4 units  (exactly √29)***

*Across 5, down 2: $c^2$ = 25 + 4 = 29.*

*Scoring: 1 point.*

**10.** Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).




**a.** Find the length of side JK. *1 pt*

***14 units***

*Across zero: 6 + 8 — never 8 − 6.*

**b.** Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

***33.9 units***

*KL = 5; JL = √(196 + 25) = √221 ≈ 14.866; 14 + 5 + 14.866 = 33.866 — round once, at the end. Follow-through on part (a).*

*Scoring: a 1 · b 1 = 2 points.*

**11.** On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

**a.** How far is it from the library to the pool along the streets? *1 pt*

***21 blocks***

*The streets follow the grid: 9 across and 12 up.*

**b.** How far is it in a straight line? *1 pt*

***15 blocks***

*The hypotenuse of legs 9 and 12: 81 + 144 = 225.*

*Scoring: a 1 · b 1 = 2 points.*

## Part 4. Similar Triangles

*3 points MA.8.GR.2.4*

**12.** The two triangles are similar. Find x. *1 pt*

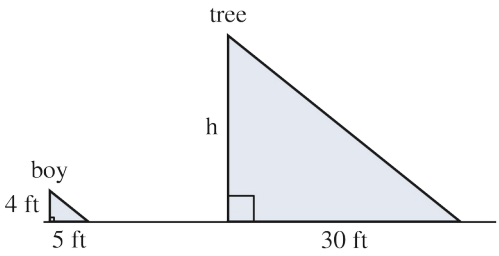



***x = 26***

*Match by size: 12 with 6 and 18 with 9, so k = 2 and x = 13 × 2. 39 matched by place on the page (18 with the 6); 13 took similar as equal; 19 added the difference 12 − 6.*

*Scoring: 1 point.*

**13.** A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*




***24 ft***

*k = 30 ÷ 5 = 6 from the shadows; 4 × 6. 37.5 matched height and shadow crosswise; 120 multiplied by the boy's height, not by k.*

*Scoring: 1 point.*

**14.** Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ● **A. 12, 14, 20**
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

***A***

*Every side × 2: 12/6 = 14/7 = 20/10.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.8.GR.1.3 | 1, 2, 3 | 3 | |
| MA.8.GR.1.1 | 4, 5, 6, 7 | 6 | |

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.8.GR.1.2 | 8, 9, 10, 11 | 6 | |
| MA.8.GR.2.4 | 12, 13, 14 | 3 | |
| **Total** | | **18** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*